SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CSA37 – SOFTWARE TESTING

EXPERIMENT 9

Case Study: Banking Application – Negative Test Scenarios and Test Cases

AIM:

To identify and prepare negative test scenarios and test cases for the Banking Application to verify that the system properly handles invalid login credentials, incorrect OTPs, insufficient account balances, invalid beneficiary details, duplicate transactions, and unauthorized access while ensuring transaction security, data integrity, and reliable banking operations.

1. Problem Description

Negative testing checks invalid credentials, invalid transfer values, insufficient balances, incorrect OTPs, duplicate submissions and unauthorized access. Transactions must not proceed when validation fails.

2. Functional Requirements

1. Customers must authenticate before accessing accounts.
2. Account balances and statements must be displayed accurately.
3. Transfers require valid beneficiaries and valid amounts.
4. Transfers cannot exceed the available balance.
5. Sensitive transactions require required verification such as OTP.
6. The system must prevent duplicate transactions and protect session data.
7. The system must provide transaction references and appropriate error messages.

3. Negative Test Cases

Negative testing verifies that invalid, incomplete, or unexpected inputs are rejected safely with suitable messages.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
NT01	Login with wrong password	Enter correct user ID and incorrect password	Access denied with safe error message	Pass
NT02	Submit blank login	Leave user ID and password empty	Required-field messages displayed	Pass
NT03	Transfer amount zero	Enter transfer amount 0	Transfer rejected	Pass
NT04	Transfer amount above balance	Enter amount greater than available balance	Transfer rejected; balance unchanged	Pass

NT05	Use invalid beneficiary account	Enter malformed or nonexistent beneficiary details	Beneficiary validation error displayed	Pass
NT06	Enter invalid OTP	Enter incorrect or expired OTP	Transaction not authorized	Pass
NT07	Repeat transfer submission	Refresh or double-click submit after a transfer	Duplicate transfer prevented or safely identified	Pass
NT08	Access account without login	Open protected account URL while logged out	Redirected to login; account data not shown	Pass
NT09	Use expired session	Attempt a transfer after session expiry	Re-authentication required; no transfer occurs	Pass
NT10	Enter invalid transfer amount	Enter letters or a negative amount	Invalid amount rejected	Pass

4. Detailed Execution of a Failed Test Case

Example: Select a test case that fails during actual execution. Record the precondition, steps, expected result, and actual result. Do not mark a test as failed unless the defect is observed.

5. Sample Bug Report

Field	**Details**
Bug ID	BR-001
Linked Test Case	Relevant failed negative test case
Severity / Priority	Assign according to impact and urgency
Expected Result	Invalid operation is rejected with a clear message
Actual Result	Application accepts or processes the invalid operation
Suggested Fix	Add server-side validation and repeat checks before final processing
Status	Open – awaiting developer investigation and correction

Conclusion

The negative test cases help verify that the Banking Application meets its requirements and responds appropriately to the tested conditions.